

Sundial × Charms Alchemy

A Cardano-Native Bitcoin Treasury Protocol & BTCfi Infrastructure Layer

Field	Detail
Proposal	Treasury Proposal for Cardano Governance
Lead Proposer	Sundial Protocol
Technical Partner	Charms
Requested Funding	10,000,000 ADA (~\$2.0M USD at reference rate of \$0.20/ADA)
Funding Split	~\$1.0M protocol infrastructure & launch liquidity · ~\$1.0M delivery, audit, integrations & go-to-market

This proposal is drafted in accordance with the Cardano Constitution v2.4 (January 24, 2026)

No DRep funds or personal compensation to Sundial or Charms principals are included in this proposal. All funding is directed exclusively toward protocol infrastructure, security audit, launch liquidity, and ecosystem integration.

At a Glance

Alchemy is a Bitcoin treasury protocol and BTCfi infrastructure layer for Cardano. It provides the reusable reserve architecture, asset primitives, and on-chain plumbing that enable Bitcoin-native finance on Cardano - delivered through a reference product (FIRE/ICE) that demonstrates the infrastructure in action.

What it builds	An on-chain BTC reserve protocol with two financial primitives - FIRE (BTC+) and ICE (BTC-) - plus open dashboards, public reporting, and integration adapters that any Cardano project can build on top of
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Why now	Strategy's MSTR trades \$2.6B/day. Ethereum's Apyx holds \$530M TVL. Structured BTC exposure is one of the fastest-growing categories in finance - and none of it is happening on Cardano
Why Cardano	Cardano's UTXO architecture makes it one of the most natural ecosystems for Bitcoin-native finance. Alchemy gives Cardano the chance to host the transparent, on-chain version of this infrastructure before it consolidates elsewhere
What the Treasury supports	Protocol development, security audit, integration adapters, public dashboards, staged launch liquidity, and go-to-market execution - with a rollover commitment that ensures no double-dipping if external investment is received
What the ecosystem gets	A reusable BTCfi infrastructure layer (open protocol, composable assets, public reporting) that wallets, DEXs, and future applications can build around - plus a live reference product bringing net new Bitcoin capital into Cardano

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The Opportunity: Bitcoin Infrastructure on Cardano

Bitcoin is the world's reserve asset of the digital age. The most attractive and sustainable crypto products are those that enhance Bitcoin's role as a store of value, reserve collateral, and provide alternative forms of exposure.

Structured Bitcoin exposure is one of the fastest-growing categories in finance:

Product	Daily Volume / TVL	What It Proves
Strategy MSTR	\$2.6B daily volume	Institutional demand for leveraged BTC exposure
Strategy STRC	\$350M daily volume (11 months old)	Demand for BTC-backed yield instruments
Strive SATA	\$100M+ volume, daily dividends	Room to compete with Strategy; daily dividends are a market first

Product	Daily Volume / TVL	What It Proves
Ethereum Apyx	\$530M TVL	On-chain BTC-backed stablecoins from corporate dividends
Ethereum/BNB Saturn	\$246M TVL	Rival protocols consolidating on EVM chains
Pendle	Holds most of Apyx's TVL	Yield-trading marketplace as complementary layer

Strategy has accumulated over 843,000 BTC at an acquisition cost of \$63.8B, proving that demand for BTC-backed senior and junior financial products is real, proven, and growing.

Yet none of this activity is happening on Cardano - despite Cardano's UTXO architecture making it one of the most natural ecosystems for Bitcoin-native finance.

If Cardano wants Bitcoin liquidity to become a real DeFi input rather than a narrative, it needs the infrastructure layer first: reserve architecture, asset primitives, safety mechanisms, integration adapters, and public reporting. Alchemy builds that layer, and delivers a reference product on top of it to prove it works.

What Alchemy Builds - The Infrastructure Layer

Alchemy is not just a product. It is a **BTCfi infrastructure layer** for Cardano - a reusable protocol that other projects, wallets, and DEXs can integrate and build on top of.

The infrastructure layer consists of three components:

1. Reserve Architecture & Safety Mechanisms

An on-chain BTC reserve system with formulaic mint/redeem constraints, dynamic reserve ratio thresholds, and transparent safety zones. This is the plumbing that makes Bitcoin-backed assets work on Cardano - and it is designed to be reused by future BTCfi protocols, not just Alchemy.

- Reserve ratio formula: $(V \times P) / L$
- FIRE price formula: $((V \times P) - L) / N^+$
- Safety zones with constrained minting/redemption at key thresholds
- All parameters public, auditable, and reported to governance

2. Composable Assets - FIRE and ICE

On top of the reserve architecture, Alchemy issues two Cardano-native assets:

FIRE (BTC+) - High-Temperature Bitcoin

FIRE makes Bitcoin **productive**. It's designed for holders who want greater exposure to Bitcoin's long-term upside.

- Receives the extra gains from the shared BTC reserve after ICE commitments are met
- Creates **amplified BTC exposure without liquidation mechanics or margin calls**
- In exchange, FIRE holders take on more short-term volatility

ICE (BTC-) - Low-Temperature Bitcoin

ICE makes Bitcoin **predictable**. It's designed for holders who want BTC-backed earning potential with more stability.

- USD-denominated, BTC-backed asset whose redemption value grows gradually over time
- **Downside buffer provided by FIRE holders**
- For users who want to benefit from Bitcoin's long-term growth without holding a highly volatile asset

Both FIRE and ICE are backed by the same shared BTC reserve. They are designed as **financial primitives** - not just products - that any Cardano application can integrate, compose, and build new products on top of.

Validated by Cardano's Own Architecture

This architecture is highly reminiscent of Cardano's own **DJED stablecoin** - but with two key differences.:

Alchemy	DJED System	Key Difference
FIRE ↔ SHEN	Volatility-absorbing claim	FIRE is backed by BTC, not ADA
ICE ↔ DJED	USD-denominated stability	ICE grows past \$1; DJED stays at \$1

These two differences give Alchemy far more appeal, virality, and staying power - reminiscent of Strategy's explosive STRC preferred stock and MSTR common stock, now worth over \$56 billion cumulatively.

3. Integration Adapters & Public Infrastructure

Beyond the protocol and asset primitives, Alchemy delivers open infrastructure that the Cardano ecosystem can build on:

- **Open Source SDKs** for wallets, DEXs, and DeFi protocols to plug into FIRE/ICE
- **Always-on public dashboards** showing reserve ratio, asset supply, liquidity health, and fee flows
- **Monthly governance reports** for DReps and the community
- **Open reporting standards** that future BTCfi projects can adopt

This BTCfi infrastructure does not exist on Cardano today. Alchemy builds it once, and the entire ecosystem benefits.

Built on Charms

Alchemy is built by Charms Inc. on the **Charms protocol** - a zero-knowledge meta-protocol that enables Bitcoin apps and assets to be natively compatible with Cardano. Charms allows FIRE and ICE to be issued on Bitcoin and circulate as Cardano-native assets from day one, plugging directly into Cardano wallets, DEXs, and DeFi tooling.

Charms credentials: Backed by Draper Associates & Draper Dragon · Open-source protocol · Bitcoin-native DEX · Live Cardano ↔ Bitcoin bridge (eBTC) · 3,000+ wallet users

How It Works - Protocol Mechanics

Alchemy's reserve architecture is organized around three core relationships:

Mechanic	Formula	Meaning
Reserve Ratio	$(V \times P) / L$	BTC reserve value vs. outstanding ICE liabilities
ICE	Senior claim	Lower-volatility, USD-denominated BTC-backed asset with formulaic growth
FIRE Price	$((V \times P) - L) / N^*$	Residual value after ICE liabilities - amplified upside, first-loss downside

Where:

Variable	Definition
V	BTC in the vault
P	BTC/USD price
L	Total ICE liability in USD
N ⁺	Total FIRE supply

The initial target reserve ratio is **5.0x**. Reserve thresholds and redemption constraints will be refined with Charms, auditors, and economic modeling before launch.

Reserve Ratio Safety Zones

Zone	Reserve Ratio	Operating Posture
Healthy	Above 4.0x	FIRE and ICE minting and redemption enabled; ICE growth continues under formula
Buffer	2.0x–4.0x	New ICE minting constrained; FIRE redemption constrained; system dynamically balances incentive mechanisms based on FIRE/ICE ratio and BTC price
Locked	Below 2.0x	FIRE minting and redemption constrained; ICE minting and redemption constrained; new risky activity paused to protect reserve integrity

Hard constraints:

- FIRE redemption is constrained below a 4.0x reserve ratio
- ICE minting is constrained below a 4.0x reserve ratio
- FIRE minting is constrained below a 2.0x reserve ratio
- ICE redemption is constrained below a 2.0x reserve ratio
- These constraints are designed to ebb and flow with the reserve ratio, creating natural incentive mechanisms that balance the system. If the ratio shifts in one direction, counterbalancing constraints are placed on the other

These constraints prevent overleveraging that puts ICE holders at risk and prevent value-destructive dilution for FIRE holders during stressed conditions.

Try the simulator: <https://www.sundialprotocol.com/alchemy/visualizer>

What Cardano Gets

Infrastructure:

- An on-chain BTC reserve protocol with composable assets (FIRE/ICE) that any Cardano wallet, DEX, or protocol can integrate and build on top of
- Open source SDKs for composability across the Cardano DeFi ecosystem
- Always-on dashboards and monthly governance reports covering reserve ratio, asset supply, liquidity health, fee flows, integrations, and milestone status
- Public reporting standards that future BTCfi projects can adopt

Reference Product:

- A live BTCfi reference product demonstrating the infrastructure in action - FIRE and ICE as Cardano-native assets with on-chain reserves and public reporting from day one, bringing net new liquidity from Bitcoin holders into the ecosystem

Treasury Yield:

- All returns generated by the launch liquidity position belong to the Cardano Treasury

Capital Efficiency:

- Sundial plans to source institutional and ecosystem-aligned capital alongside the Treasury allocation, so Treasury support catalyzes a larger BTCfi liquidity base flowing to Cardano rather than carrying the market alone
- The rollover commitment ensures that if external investment is received, development funding rolls into additional liquidity rather than duplicating compensation

Why Cardano Needs This Now

Structured Bitcoin exposure is one of the fastest-growing categories in finance. Through this vertical, Cardano has a once-in-a-generation opportunity to claim the on-chain BTCfi category - before it consolidates on Ethereum, BNB, or elsewhere.

Strategy's MSTR and STRC products — leveraged equity and USD-denominated yield instruments backed by a Bitcoin treasury — have become some of the most actively traded instruments in public markets. MSTR trades for \$2.6 billion in volume per day, while STRC trades \$350M per day at just 11 months old. Through these instruments, Strategy has

accumulated over 843,000 BTC on its balance sheet at an acquisition cost of \$63.8 billion, proving the demand for BTC-backed senior and junior financial products is real, proven, and growing.

It's not just Strategy either. Strive, a direct competitor to Strategy, accumulated over 4,600 BTC in the last week of May using its Bitcoin-backed preferred stock, SATA, after debuting daily dividends for the first time in stock market history, outperforming Strategy. With over \$100M in trading volume, Strive proves there is room to compete with Saylor in this product category (in fact, Saylor often provides free mass marketing for his competitors just to advance the mission!).

Yet none of that activity is happening on Cardano, despite Cardano's UTXO architecture making it one of the most natural ecosystems for Bitcoin-native finance.

If anything, this leg of BTCfi is already expanding rapidly on Ethereum. Apyx is a protocol that issues stablecoins on Ethereum, backed by off-chain corporate dividends from Bitcoin Treasury Company preferred equities, and has already gathered \$530M TVL. Saturn, a rival protocol on Ethereum and BNB, has a \$246M TVL. Pendl, a complementary yield-trading marketplace that lets users separate and speculate on future yields of these stocks, holds most of Apyx's TVL, whose assets are the most popular on its platform.

If Cardano wants Bitcoin liquidity to become a real DeFi input rather than a narrative, it needs credible on-chain infrastructure now, before the category is settled elsewhere. That means reserve architecture, asset primitives, safety mechanisms, open source SDKs, and public reporting - before products can be built on top of it.

Alchemy builds that infrastructure, and delivers a reference product on top to prove it works. This is how ecosystems grow: **infrastructure first, then products, then network effects**. Cardano has the architecture. Alchemy provides the BTCfi layer that makes it useful.

The economics and demand are validated by Strategy and its tokenization layers. The on-chain reserve architecture is validated by Cardano's own DJED stablecoin, which has run with formally verified safety properties for years using the same senior/junior two-token model.

For the first time, Cardano can host the transparent, programmable version of an emerging, killer product category **before** other ecosystems. There is immense room to compete, and we must seize the opportunity.

Requested Funding

Total ask: 10,000,000 ADA (~\$2.0M USD at reference rate of \$0.20/ADA)

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The request is divided into two clearly separated pools:

Pool 1: ~\$1.0M Protocol Infrastructure & Launch Liquidity

This is the heart of the proposal. The infrastructure and liquidity pool is treasury-supported, not general operating capital. It funds the protocol's core infrastructure deployment and staged liquidity seeding, kept separate from delivery spend, and governed by reporting and use-restriction rules.

Treasury-supported liquidity is not risk-free. Once deployed, it may be exposed to defined protocol, market, oracle, bridge/locker, and liquidity risks. The proposal limits deployment by tranche, requires audit and reporting gates, and keeps liquidity use separate from Sundial and Charms operating costs.

Purpose:

- Deploy and seed the on-chain BTC reserve protocol and its safety mechanisms
- Seed the shared BTC reserve to make the Alchemy launch credible with real liquidity depth
- Demonstrate the mechanics of FIRE and ICE with initial capital
- Provide Cardano with visible BTCfi infrastructure and a live reference product
- Help establish Cardano as the infrastructure layer for transparent, on-chain BTC-backed structured exposure

Staged Deployment:

Stage	Amount	Release Condition
Month 1	\$250K	After audit and launch-readiness review
Month 2	\$250K	After public reporting, operational review, and 30-day grace period
Month 3	\$500K	After mint/redeem thresholds, reserve ratio tracking, growth rate monitoring, and dashboard performance are confirmed

Initial Reserve Target:

Side	Amount
FIRE-side liquidity	~\$800K
ICE-side liquidity	~\$200K

Final allocation between FIRE and ICE will be optimized to maintain a 5.0x reserve ratio, adjusted based on economic modeling, audit feedback, and launch conditions.

Liquidity Controls & Reporting:

- Treasury-supported launch liquidity is deployed only under approved tranche, reporting, and use-restriction rules
- Launch liquidity will be **owned solely by the Treasury**. All profits, yield, and returns will be returned to the Treasury on a quarterly basis, converted from their original asset into ADA via Cardano-native DEXs at commercially reasonable rates, with conversion timing and methodology disclosed in advance in monthly reports
- Each tranche purchased at a 4:1 ratio of FIRE to ICE to push the reserve ratio towards 5x
- The principal can be returned to the Treasury after Alchemy's protocol TVL > \$60M, via formal governance proposal for DRep and Constitutional Committee vote. If the governance vote does not pass, the principal remains in the liquidity position and the proposal will be resubmitted at the next governance cycle. ~~£~~\$10,000,000 return proposal condition: Upon Alchemy achieving a 30-day time-weighted average Total Value Locked of US\$60,000,000 or above, measured by on-chain reserve ratio × BTC/USD Chainlink oracle data, with weekly attestation by the fund administrator and annual independent audit confirmation.
- All liquidity deployment is staged with third-party administration. **Intersect** is proposed as fund administrator on an interim basis, subject to confirmation and final agreement. Fund administration and oversight arrangements may be adjusted as governance infrastructure evolves
- Yield generated returns to the Cardano Treasury

ADA Price Protection: The maximum acceptable ADA reference rate is **\$0.35**. If ADA appreciates above this before or during delivery, future milestone withdrawals will be reduced or excess ADA will be returned. The Treasury funds the work, not a windfall.

If milestones are missed, reporting lapses, or liquidity-health thresholds are breached, further deployment pauses until issues are resolved or governance provides direction.

Pool 2: ~\$1.0M Delivery, Audit, Integrations & Go-to-Market Budget

This funds the work required to design, build, audit, launch, integrate, and report on Alchemy - including the protocol infrastructure, composability adapters, dashboards, and ecosystem integration that make it reusable beyond a single product.

Allocation	Workstream	Scope
\$250,000	Protocol infrastructure	Charms Scrolls infrastructure, BTC asset logic, reserve mechanics, mint/redeem constraints, upgrade support
\$300,000	Engineering, platform & delivery	Product architecture, platform integration, institutional frontend/backend, monitoring, reporting, capital-formation support
\$200,000	Security review, audit & economic modeling	Independent review of protocol logic, reserve thresholds, smart-contract-like execution, oracle dependencies, economic assumptions
\$100,000	Public dashboards, UX & DeFi composability	Public-facing dashboards, wallet flows, DEX integration adapters, developer documentation, composability guides for ecosystem projects
\$75,000	Legal, compliance, disclosures & risk documentation	ICE/FIRE positioning, disclosures, treasury reporting structure, risk documentation, launch-readiness review
\$50,000	Ecosystem education, governance reporting & launch	Governance communications, DRep education, ecosystem partner activation, launch coordination
\$25,000	Fund administration & accounting	Intersect or equivalent administration and reporting support

These are planning allocations within a fixed ~\$1.0M delivery budget. Funds may be reallocated between categories as requirements are finalized, but total delivery spend will not exceed the approved budget. The infrastructure and launch liquidity pool remains separate and cannot be used for implementation costs, delivery overruns, operating expenses, or any purpose outside approved deployment. Material reallocations will be disclosed in milestone reporting.

Rollover Commitment

Sundial and Charms commit to the following: **if either Sundial or Charms receives any investment for Alchemy development, the funding portion of this proposal that is allocated to the development of Alchemy will instead be rolled over into additional treasury-supported launch liquidity.**

This means:

- The delivery work still gets done - funded by external investment instead of the Treasury
- The Treasury gets **more liquidity depth without additional cost**
- There is **no double-dipping**: we will not pocket Treasury funds for development work that is already funded by external investment

This commitment ensures that Treasury funding is additive, not duplicative. If outside capital accelerates the project, the Treasury benefits from greater liquidity rather than subsidizing work that's already funded.

Fund Administration & Treasury Controls

Sundial is accountable for controlled deployment of treasury-linked funds, reporting, and compliance with the proposal's use restrictions.

Intersect is proposed as professional fund administrator on an interim basis, subject to confirmation and final agreement. Should Intersect not confirm within 30 days of governance enactment, the proposer shall nominate an alternative independent fund administrator, subject to Constitutional Committee review. The designated fund administrator (currently proposed as Intersect, subject to confirmation) shall delegate any ADA associated with this proposal to an auto-abstain stake credential, not to a Stake Pool Operator, per Appendix I, Section 5.

Intersect would support fund administration, accounting, and oversight, including:

- Defined use restrictions
- Reserve status reporting
- Deployment pause and review rules for underperforming or non-compliant liquidity
- Treasury-facing reporting on reserve status, liquidity health, integrations, incidents, and milestone progress

Fund administration arrangements may be adjusted as Cardano's governance infrastructure evolves. Any change to fund administrator shall be proposed via the same dispute resolution

mechanism or a new governance action, with minimum 14-day community notice period and Constitutional Committee review.

Always-on dashboards and monthly reports will be published for DReps and the community, including deployed liquidity, reserve ratio, FIRE/ICE supply, liquidity health, incidents, material parameter changes, integrations, go-to-market progress, and upcoming milestone conditions.

Sundial will disclose any related-party relationships or material commercial arrangements involving Sundial, Charms, or any third parties connected to this proposal.

Who's Building It

Sundial Protocol - Lead Proposer

Bitcoin-native credit and derivatives platform.

- 1st place, Institutional Track - Paris Blockchain Week (3rd overall out of 2,000+)
- Hacken-audited BTC platform live in testnet
- Currently completing Fund 13 Catalyst scope - delivering core Bitcoin-on-Cardano infrastructure, including architecture design, protocol development, security validation, and institutional DeFi components

Sundial brings product architecture, institutional capital formation, go-to-market execution, and accountability for treasury-linked funds.

Charms - Core Technical Partner

Bitcoin meta-protocol specialist.

- Backed by Draper Associates & Draper Dragon
- Open-source protocol
- Bitcoin-native DEX
- Live Cardano ↔ Bitcoin bridge (eBTC)
- 3,000+ wallet users

Charms designs the Alchemy protocol layer and core FIRE/ICE mechanics.

Risks

Risk Category	Nature	Mitigation
BTC Price Risk	Treasury-supported liquidity is exposed to BTC price risk once deployed	Automatic reserve thresholds limit cascading drawdowns; not eliminated
Oracle & Bridge Dependencies	Technical risk from external infrastructure	Independent security audit funded by this proposal; Sundial's Hacken-audited infrastructure as starting point. Charms protocol layer risk: Alchemy relies on Charms Scrolls and beaming infrastructure for Bitcoin-native asset issuance and Cardano-native circulation. While Charms is production software with live bridge activity, any protocol-level vulnerability in Scrolls or beaming could affect Alchemy's ability to mint, redeem, or accurately account for reserve-backed assets. Mitigation: (a) \$200,000 independent security review of Charms Scrolls and beaming smart contracts, funded through this proposal; (b) contract upgrade and patch continuity plan, ensuring Alchemy's reserve architecture can be maintained and patched in-place via standard Cardano smart contract upgrade paths without requiring third-party custody migration
Novel Instruments	FIRE and ICE are new asset types	Legal and regulatory analysis before mainnet; no communication will describe ICE as risk-free until legal structure supports it
Delivery Risk	Scope or timeline overruns	Milestone-gated structure; Treasury exposure scales with demonstrated delivery

In the event of disputes regarding milestone assessment, fund release conditions, or administrator actions, the matter shall be referred to an independent reviewer or the Constitutional Committee for resolution.

The Bottom Line

Alchemy is not a speculative experiment. It is a **proven product category - BTC-backed structured exposure - rebuilt as on-chain infrastructure: transparent, composable, globally accessible, and integrated with Cardano DeFi from day one.**

The demand is validated by Strategy's \$63.8B BTC treasury and \$2.6B daily MSTR volume. The architecture is validated by Cardano's own DJED stablecoin running for years with the same two-token model. The timing is validated by Apyx, Saturn, and Pendle, consolidating this category on Ethereum right now.

Cardano has the architecture. The market has the demand. Alchemy builds the infrastructure that connects them.

10,000,000 ADA in targeted, milestone-gated funding delivers:

- An on-chain BTCfi infrastructure layer - reserve architecture, composable BTC-backed asset primitives, public dashboards - that any Cardano project can build on top of
- Two Cardano-native assets (FIRE and ICE) with transparent on-chain reserves and governance reporting
- A live reference product demonstrating the infrastructure in action, bringing net new BTC capital into Cardano
- Treasury-owned launch liquidity with quarterly yield returns
- A rollover commitment ensuring no double-dipping
- DRep-visible dashboards and monthly reporting
- Institutional go-to-market execution bringing net new BTC capital into Cardano

This is a focused strategic investment in Cardano's Bitcoin infrastructure - not a broad subsidy, not a general operating grant, and not private balance-sheet support.

This is the moment. Cardano can build the BTCfi infrastructure layer, or watch it settle elsewhere.